Jackson Bean
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Education

B.S. in Computer Science and Engineering **August 2021 – May 2026**
The Ohio State University
Double major with Theoretical Mathematics
Specialization in Software Engineering
GPA: 4.00/4.00

Research

Undergraduate Research Mentee **September 2025 – April 2026**
Cycle Program, The Ohio State University *Advisors: Levi Keck and Torey Hilbert*

- Analyzed symplectic, area-preserving numerical methods used to simulate interactions between two Argon atoms.
- Studied modified and computational differential equations to represent complex, real-world Hamiltonian systems.
- Developed a Python simulation which measured the energy conservation of general numerical methods against symplectic methods.

Undergraduate Researcher **May 2024 – September 2024**
Cognitive Systems and Engineering Lab, The Ohio State University *Advisor: Dr. Martijn IJtsma*

- Integrated two new decision-making contingency plans designed for UAMs/UAVs in an urban airspace.
- Developed C++ code to run 50 simulations that enacted parameter-determined decision-making.
- Analyzed results of simulations using pandas and Pyplot Dash in Python to create six dynamic charts and graphs.

Undergraduate Researcher **May 2023 – August 2023**
Department of Biomedical Informatics, The Ohio State University *Advisor: Dr. Ping Zhang*

- Constructed an AI model trained using gradient boosting, random forests, and SVMs with scikit-learn in Python that classified patient outcomes with more than 80% accuracy.
- Utilized pandas and NumPy to efficiently access over one million instances of patient data for training.
- Compared over 150 instances of training and test results using matplotlib in Python and presented findings at a departmental research forum.

Teaching

Student Instructional Associate **September 2024 – May 2026**
Department of Mathematics, The Ohio State University

MATH 1130 - College Algebra for Business: Fall 2024, Spring 2025

MATH 1148 - College Algebra: Fall 2025, Spring 2026

- Prepared and taught recitation twice a week to over 70 students into two sections.
- Answered over 20 questions each week about course material in a concise and understandable manner.
- Graded and proctored 10 weekly quizzes and four monthly exams over semester-long course content.

Skills

- **Languages**: Python, SQL, C/C++, MATLAB, Java, LaTeX, HTML/CSS, R, JavaScript, Ruby
- **Frameworks/Libraries**: NumPy, pandas, scikit-learn, React, Tailwind CSS, Express.js, Ruby on Rails
- **Developer Tools**: MongoDB, SQLite, Linux/UNIX, Makefile, Git, Vite, VS Code, Xcode, Docker
- **Miscellaneous**: Personal Computer Building, Global Seal of Biliteracy in Spanish

Relevant Coursework

Mathematics and Statistics Courses

- MATH 5801: General Topology and Knot Theory — Spring 2026
- MATH 4581: Abstract Algebra II — Spring 2026
- MATH 4557: Partial Differential Equations — Spring 2026
- MATH 4580: Abstract Algebra I — Fall 2025
- MATH 4548: Introductory Analysis II — Spring 2025
- STAT 4202: Introduction to Mathematical Statistics II — Spring 2025
- MATH 4547: Introductory Analysis I — Fall 2024
- STAT 4201: Introduction to Mathematical Statistics I — Fall 2024
- MATH 3345H: Honors Foundations of Higher Mathematics — Spring 2024
- MATH 3607: Beginning Scientific Computing — Fall 2023
- MATH 2415: Ordinary and Partial Differential Equations — Spring 2022
- MATH 2568: Linear Algebra — Fall 2021

Computer Science Courses

- CSE 5914: Capstone Design: Knowledge-Based Systems — Spring 2026
- CSE 3321: Automata and Formal Languages — Spring 2026
- CSE 3461: Computer Networking and Internet Technologies — Fall 2025
- CSE 3341: Principles of Programming Languages — Fall 2025
- CSE 3241: Introduction to Database Systems — Fall 2025
- CSE 3232: Software Requirements Analysis — Fall 2025
- CSE 3231: Software Engineering Techniques — Fall 2025
- CSE 3901: Project: Design, Development, and Documentation of Web Applications — Spring 2025
- CSE 3521: Survey of Artificial Intelligence I: Basic Techniques — Spring 2025
- CSE 2431: Systems II: Introduction to Operating Systems — Spring 2025
- CSE 2421: Systems I: Intro to Low-Level Programming and Computer Organization — Fall 2024
- CSE 2331: Foundations II: Data Structures and Algorithms — Fall 2024
- CSE 2231: Software II: Software Development and Design — Spring 2024
- CSE 2321: Foundations I: Discrete Structures — Spring 2024
- ECE 2060: Introduction to Digital Logic — Spring 2024
- CSE 2221: Software I: Software Components — Fall 2023
- ECE 2020: Introduction to Analog Systems and Circuits — Fall 2023